Prateek Awate

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Professional Summary

AI engineer with 4.5+ years across computer vision and production LLM systems. Deployed real-time vision on edge hardware, face-recognition personnel identification (InsightFace, 99%+ precision) and YOLO person detection on Raspberry Pi plus air-gapped GenAI systems (Llama 3 8B) on NVIDIA GPUs. Architected 6 production microservices serving 500+ users with 70% API cost reduction; published CLEF 2026 research ranking 1st in two competition tasks.

Experience

Senior AI Engineer | Pridenexus Pvt Ltd | Dec 2025 - Present | Pune, India

- Architected 3 AI platforms across 6 FastAPI microservices serving 500+ users React/TypeScript, PostgreSQL, Weaviate, Redis, Kafka, SSE streamin.
- Prototyping vision-language document pipeline (pilot), OpenCV preprocessing, fine-tuned YOLOv8n for region detection (localization + classification), MobileSAM segmentation for region isolation, and LoRA-fine-tuned SmolVLM for structured extraction; evaluated with precision, recall, F1, and mAP, benchmarking zero-shot vs fine-tuned.
- Cut API costs 70% via prompt engineering, multi-model routing (Gemini + Claude), token pruning, and Redis caching.
- Built async-first infrastructure (SQLAlchemy 2.0, Weaviate RAG, self-healing retries) with Google Search grounding for auditable outputs.
- Operationalized DevOps with Docker and observability (Prometheus/Grafana, LangSmith); mentored 3 engineers.

AI Engineer | Technocraft Solutions | Dec 2024 - Dec 2025 | Mumbai, India

- Architected and deployed air-gapped on-prem GenAI system (Llama 3 8B, BF16) for querying confidential documents — emails, PDFs, Word, Excel, 10KB–12MB per file — with zero external connectivity; optimized inference on NVIDIA RTX 4090.
- Built real-time personnel identification and automated attendance system: InsightFace face recognition (99%+ precision) with YOLO person detection on Raspberry Pi — OpenCV camera integration and frame processing, entry-event detection, timestamped logging, fully on-device.
- Developed ChatGPT-style UI with RAG pipeline providing direct source attribution for full auditability across the document corpus.
- Built modular, config-driven codebases enabling rapid iteration in restricted air-gapped environments.

Associate Software Engineer | Mphasis Ltd | Dec 2021 - Feb 2024 | Mumbai, India

- Built enterprise data pipelines (SQL, Pandas, NumPy) automating ingestion and transformation of large-scale datasets.
- Led end-to-end development of a data-driven web application (ReactJS, Spring Boot); 30% increase in user engagement.
- Performed database debugging, performance tuning, and schema optimization for enterprise clients.

Assistant Tech Support | Shri Girijatmak Pvt. Ltd | Apr 2021 – Dec 2021 | Pune, India

Delivered technical support to multiple clients with 25% improvement in customer satisfaction. Developed corporate website achieving 35% increase in visitor engagement. Managed 500+ critical database records and facilitated product demonstrations for prospective clients.

Core Skills

Production AI & LLM Systems: Multi-agent orchestration (LangChain, LangGraph, phase-based conversation flows) · LLM integration and routing (Gemini, Claude, Llama 3.1, intelligent model selection) · Cost optimization (prompt engineering, multi-model routing, token management, caching) · RAG systems (Weaviate, pgvector, source attribution, retrieval ranking) · Real-time inference (SSE streaming, low-latency pipelines, concurrent request handling) · Production observability (Prometheus/Grafana, LangSmith · Python.

Infrastructure & Engineering: Python (async-first, FastAPI, production-grade) · SQLAlchemy 2.0 async ORM · PostgreSQL, Redis, Kafka · Docker containerization and automation · TypeScript/React frontends · Database optimization and tuning · JWT auth, rate limiting, Alembic migrations.

ML & Research: PyTorch, TensorFlow, Scikit-learn, Transformers · Quantum computing (QUBO formulation, D-Wave optimization) · Multi-agent systems design · Data-efficient learning strategies · Edge inference optimization (NVIDIA Jetson, RTX 4090) · FAISS vector search · Bayesian optimization (BoTorch).

Selected Publications

Quantum-Accelerated Instance Selection for Sentiment Classification - CLEF 2026

Ranked **1st among pure Quantum Annealing submissions** (Task 2) and **4th overall**. Solo research: formulated training-set instance selection as novel QUBO with cluster-aware penalties, achieving 71% training-data reduction while preserving classifier quality. Only team where QA outperformed Simulated Annealing (+4.4 Macro-F1). Paper accepted at CLEF 2026 Working Notes (Jena, Germany).

Multi-Agent Swarm for Cross-Lingual Wordplay Translation - CLEF 2026

Ranked **1st overall in Task 2** (English-to-French Pun Translation), outperforming closest competing lab by +4.6 BLEU points. Co-authored 8-stage hybrid LLM pipeline with 5-channel FAISS vector search combining cross-lingual semantic and phonetic embeddings, evaluated by 4-judge "Panel of LLMs" swarm for iterative self-correction.

Co-Author: Machine Learning Applications for Predicting Cardiac Events

Published in International Research Journal of Modernization in Engineering Technology and Science (IRJMETS). Comparative analysis of supervised learning algorithms for early cardiac event detection, achieving 98% recall on 900+ patient dataset.

Research in Progress

LITE Batching: Light-Weight Training Batch Selection Using Past Training Statistics

Developing a forward-pass-free selection policy to eliminate O(N) computational overhead in gradient-based data selection methods. Utilizing inexpensive training signals (loss trajectories, forgetting events) to identify high-value training batches without per-sample gradients or auxiliary holdout models. Bridges gap between high-cost dynamic selection and low-cost static pruning, enabling efficient, architecture-agnostic training for large-scale Vision Transformers and NLP models. Targets deployment in production training pipelines where computational efficiency is critical.

Education

- **B.E. Electronics and Telecommunication**, Dr. D. Y. Patil School of Engineering, Pune (May 2021)

Certifications

- **DP-900** Microsoft Certified: Azure Data Fundamentals
- **NVIDIA** - Building LLM Applications with Prompt Engineering
- **NVIDIA** - Rapid Application Development using Large Language Models

Languages

English (Fluent) · Hindi (Native) · Marathi (Native) · Japanese (Conversational)